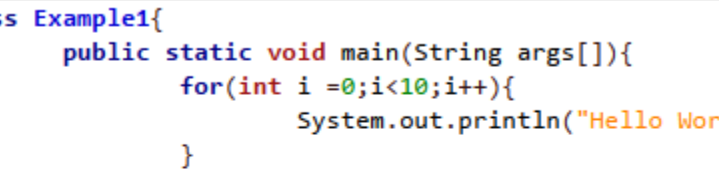


Q1.

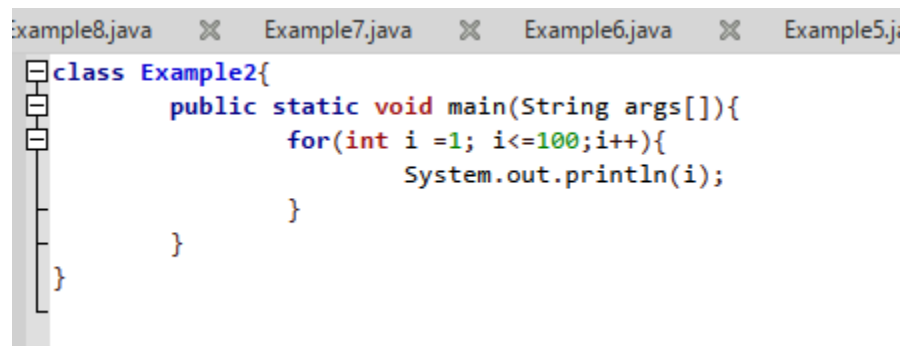


```
class Example1{
    public static void main(String args[]){
        for(int i =0;i<10;i++){
            System.out.println("Hello World");
        }
    }
}
```

Output.

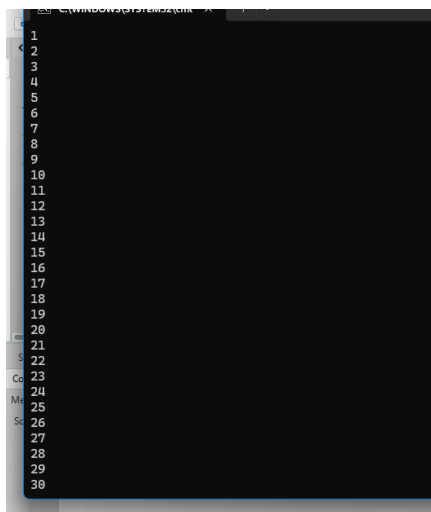
[illegible]

Q2.



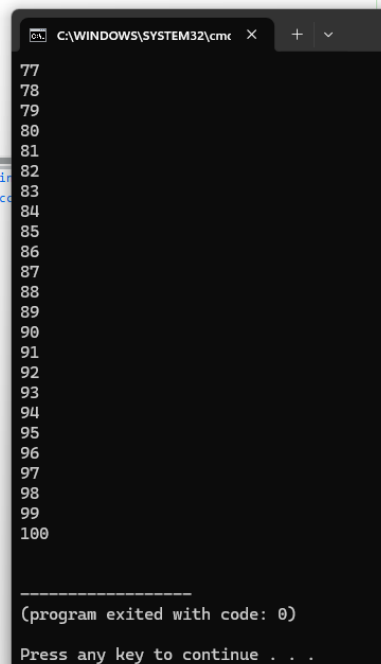
```
Example8.java  Example7.java  Example6.java  Example5.java
class Example2{
    public static void main(String args[]){
        for(int i =1; i<=100;i++){
            System.out.println(i);
        }
    }
}
```

Output.



```
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
```

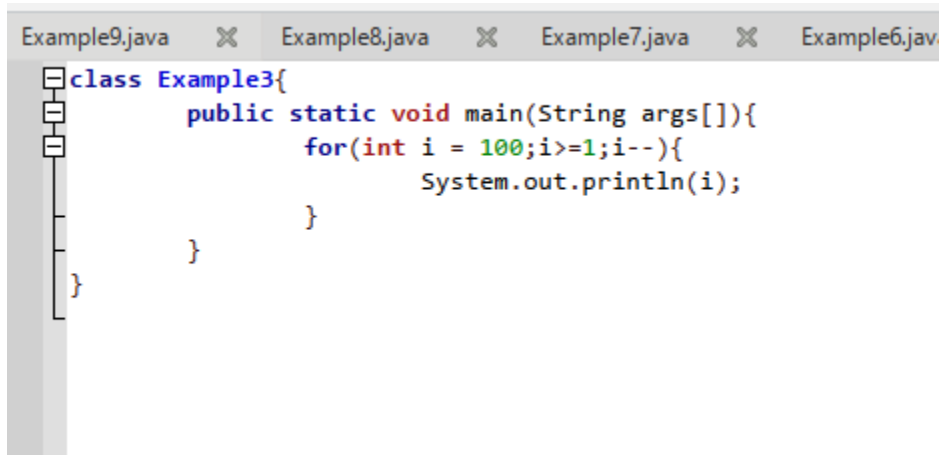
java" (in
shed succ



```
C:\WINDOWS\SYSTEM32\cmd
77
78
79
80
81
82
83
84
85
86
87
88
89
90
91
92
93
94
95
96
97
98
99
100

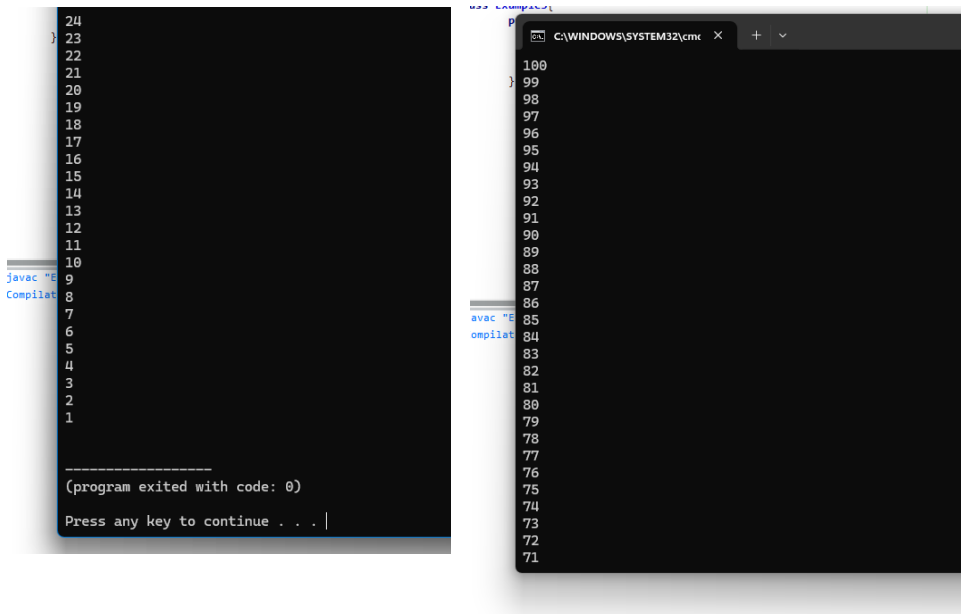
-----
(program exited with code: 0)
Press any key to continue . . .
```

Q3.



```
Example9.java  Example8.java  Example7.java  Example6.java
class Example3{
    public static void main(String args[]){
        for(int i = 100;i>=1;i--){
            System.out.println(i);
        }
    }
}
```

Output.

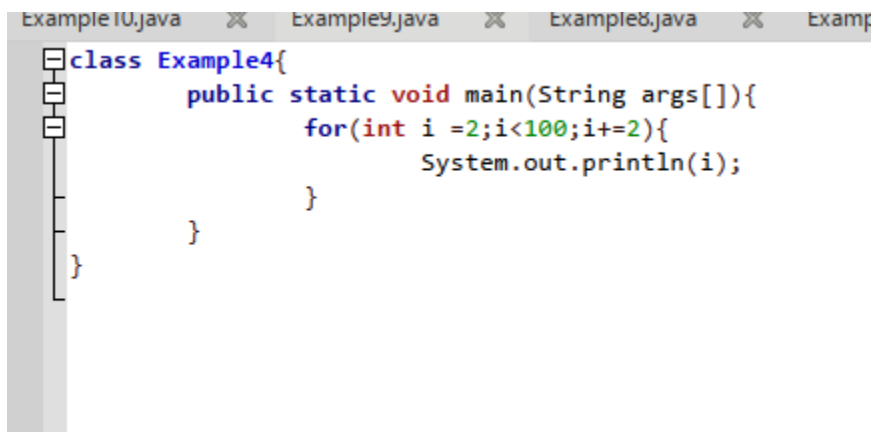


```
24
23
22
21
20
19
18
17
16
15
14
13
12
11
10
9
8
7
6
5
4
3
2
1

-----
(program exited with code: 0)
Press any key to continue . . . |

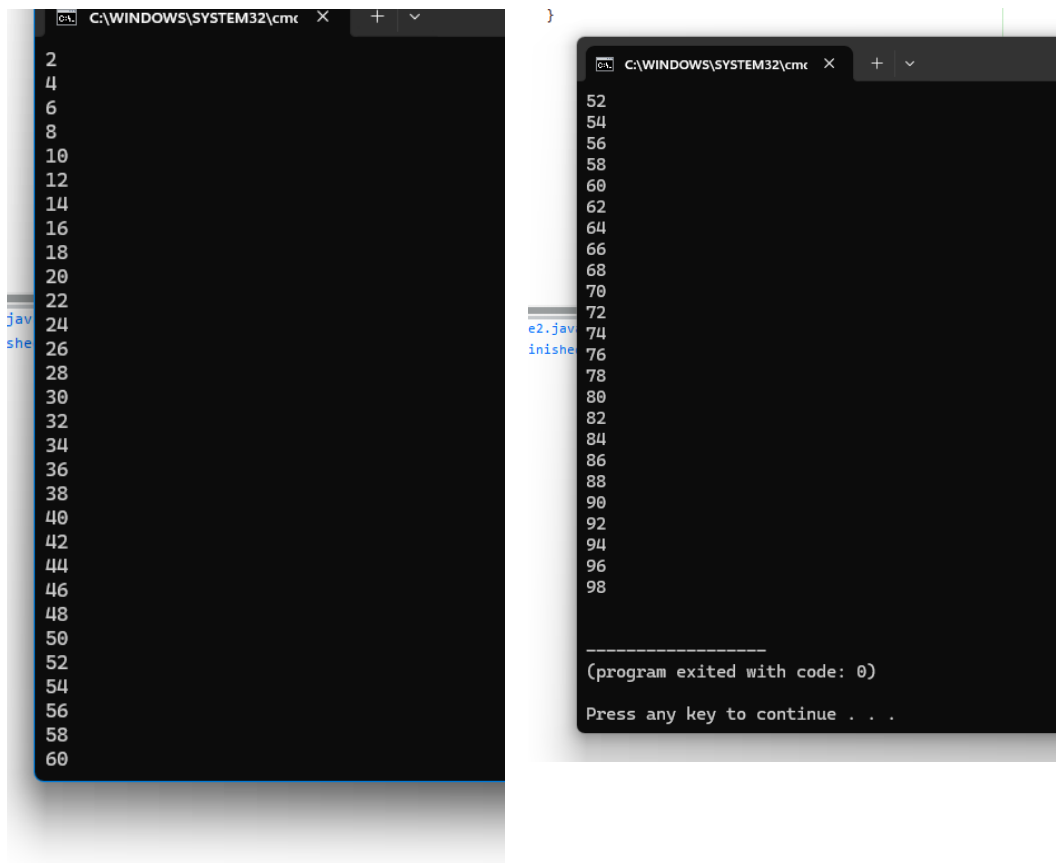
C:\WINDOWS\SYSTEM32\cmd
100
99
98
97
96
95
94
93
92
91
90
89
88
87
86
85
84
83
82
81
80
79
78
77
76
75
74
73
72
71
```

Q4.



```
Example10.java  Example9.java  Example8.java  Examp
class Example4{
    public static void main(String args[]){
        for(int i =2;i<100;i+=2){
            System.out.println(i);
        }
    }
}
```

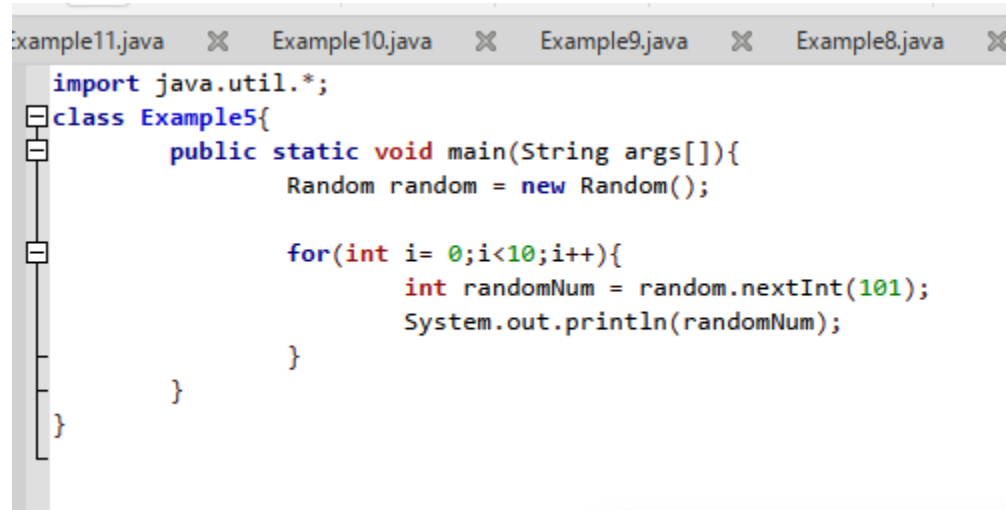
Output.



```
C:\WINDOWS\SYSTEM32\cmd  X  +  v
2
4
6
8
10
12
14
16
18
20
22
24
26
28
30
32
34
36
38
40
42
44
46
48
50
52
54
56
58
60
62
64
66
68
70
72
74
76
78
80
82
84
86
88
90
92
94
96
98

-----
(program exited with code: 0)
Press any key to continue . . .
```

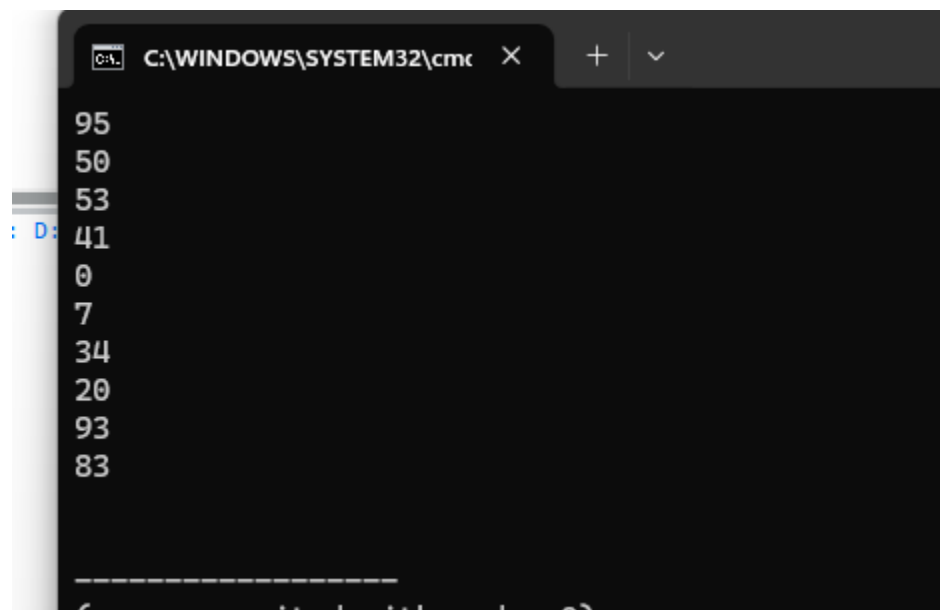
Q5.



```
Example11.java  Example10.java  Example9.java  Example8.java  Example5.java
import java.util.*;
class Example5{
    public static void main(String args[]){
        Random random = new Random();

        for(int i= 0;i<10;i++){
            int randomNum = random.nextInt(101);
            System.out.println(randomNum);
        }
    }
}
```

Output.

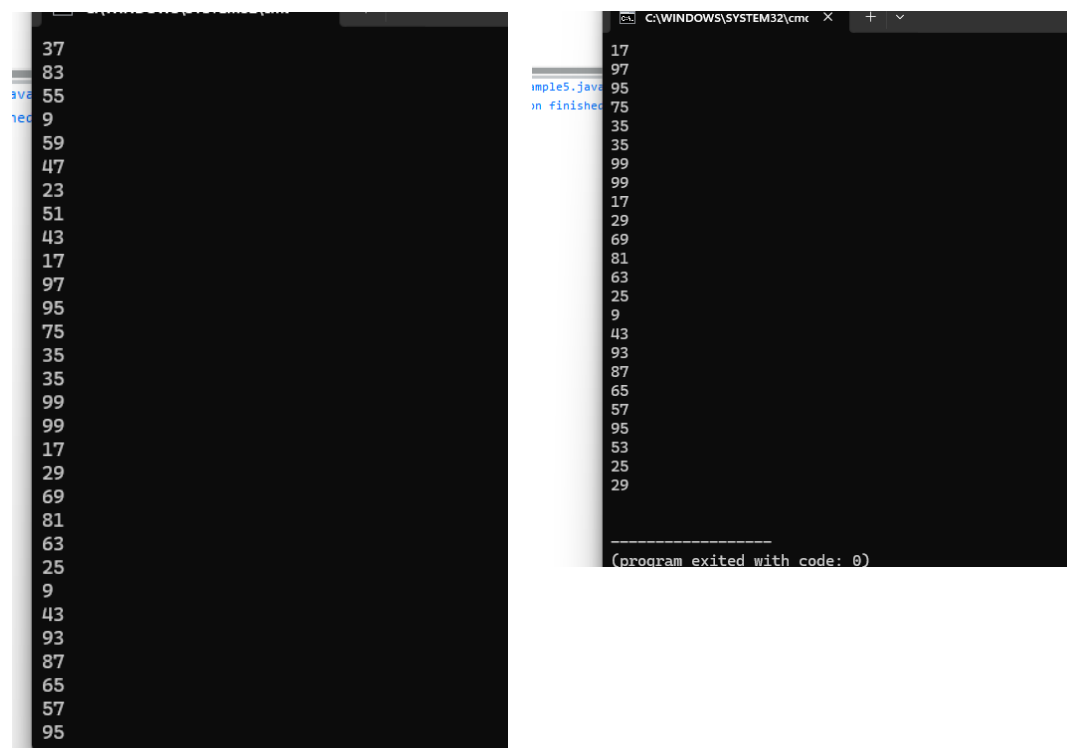


```
C:\WINDOWS\SYSTEM32\cmd.exe
95
50
53
41
0
7
34
20
93
83
-----
C:\WINDOWS\SYSTEM32\cmd.exe
```

Q6.

```
import java.util.*;
class Example6{
    public static void main(String args[]){
        Random r = new Random();
        for(int i = 1;i<=50;i++){
            int rNum = r.nextInt(101);
            if(rNum%2==1){
                System.out.println(rNum);
            }
        }
    }
}
```

Output.



37
83
55
9
59
47
23
51
43
17
97
95
75
35
35
99
99
17
29
69
81
63
25
9
43
93
87
65
57
95
53
25
29

(program exited with code: 0)

Q7.

```
class Example7{
    public static void main(String args[]){
        for(char c = 'A'; c <= 'Z'; c++){
            System.out.print(c+" ");
        }
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd X + v
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q8.

```
Example14.java  Example15.java  Example12.java  Example11.java  Example10.java
import java.util.*;
class Example8{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter number to find factorial :");
        int num = input.nextInt();

        int numPlus=0;
        int fac =1;
        for(int i = num ;i>0;i--){
            fac *= i;
        }
        System.out.print(fac);
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd
Enter number to find factorial :5
120

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q9.

```
Example15.java  X Example14.java  X Example13.java  X Example12.java  X Example11.jav
import java.util.*;
class Example9{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        int num =0;
        int x =0;
        for(int i =1;i<=50;i++){
            System.out.print("Enter the number"+i+" : ");
            num = input.nextInt();
            if(num>100){
                x++;
            }
        }
        System.out.println();
        System.out.print(x+" numbers are more than 100");
    }
}
```

Output.

```
Enter the number31 : 45
Enter the number32 : 34
Enter the number33 : 534
Enter the number34 : 53
Enter the number35 : 45
Enter the number36 : 34
Enter the number37 : 53
Enter the number38 : 453
Enter the number39 : 45
Enter the number40 : 345
Enter the number41 : 3453
Enter the number42 : 5
Enter the number43 : 35
Enter the number44 : 34
Enter the number45 : 53
Enter the number46 : 534
Enter the number47 : 5345
Enter the number48 : 345
Enter the number49 : 3454
Enter the number50 : 45

23 numbers are more than 100

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q10.

```
Example16.java  Example15.java  Example14.java  Example13.java  Example12.java  Example11
import java.util.*;
class Example10{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        int total = 0;

        int min = 0;
        int max = 0;
        for(int i =1;i<=10;i++){

            System.out.print("Enter the Subject "+i+" marks : ");
            int mark = input.nextInt();
            total += mark;
            if(i==1){
                max=mark;
                min=mark;
            }else{
                if(mark>max){
                    max=mark;
                }
                if(mark<min){
                    min=mark;
                }
            }

        }
        int avg =total/10;
        System.out.println("\nTotal    : " + total);
        System.out.println("Max      : " + max);
        System.out.println("Min      : " + min);
        System.out.println("Average : " + avg);
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd  +  v
Enter the Subject 1 marks : 56
Enter the Subject 2 marks : 57
Enter the Subject 3 marks : 98
Enter the Subject 4 marks : 67
Enter the Subject 5 marks : 87
Enter the Subject 6 marks : 65
Enter the Subject 7 marks : 87
Enter the Subject 8 marks : 45
Enter the Subject 9 marks : 78
Enter the Subject 10 marks : 90

Total    : 730
Max      : 98
Min      : 45
Average : 73

-----
(program exited with code: 0)
```

Q11.

```
Example11.java X Example17.java X Example18.java X Example19.java X nple20.java X
import java.util.*;
class Example11{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        double totalH=0;
        double totalW=0;
        for(int i =1;i<=100;i++){
            System.out.println("Student "+i);
            System.out.print("Enter Student "+i+" Height : ");
            double height =input.nextDouble();

            System.out.print("Enter Student "+i+" Weight : ");
            double weight =input.nextDouble();
            System.out.println();
            totalH += height;
            totalW += weight;

        }
        double avgH = totalH/100;
        double avgW = totalW/100;
        System.out.println("Average height is " +avgH);
        System.out.println("Average Weight is " +avgW);
    }
}
```

Output.

```
Student 96
Enter Student 96 Height : 798
Enter Student 96 Weight : 79

Student 97
Enter Student 97 Height : 79
Enter Student 97 Weight : 79
}
doub
doub
Syst
Syst
Student 98
Enter Student 98 Height : 7
Enter Student 98 Weight : 9

Student 99
Enter Student 99 Height : 987
Enter Student 99 Weight : 98

Student 100
Enter Student 100 Height : 98
Enter Student 100 Weight : 98

Average height is 258.86
Average Weight is 146.39

-----
(program exited with code: 0)

Press any key to continue . . . |
```

Q12.

```
Example10.java Example11.java Example10.java Example13.java Example14.java
class Example12{
    public static void main(String args[]){
        System.out.println("For loop");
        System.out.println();
        for(int i = 0; i<10;i++){
            System.out.println("Hello world");
        }
        System.out.println();
        System.out.println();
        //-----
        System.out.println("While loop");
        System.out.println();
        int i =1;
        while(i<=10){
            System.out.println("Hello world");
            i++;
        }
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd.exe
Hello world
Hello world
Hello world
Hello world
Hello world
Hello world
Hello world
Hello world
Hello world
Hello world

While loop

Hello world
Hello world
Hello world
Hello world
Hello world
Hello world
Hello world
Hello world
Hello world
```

Q13.

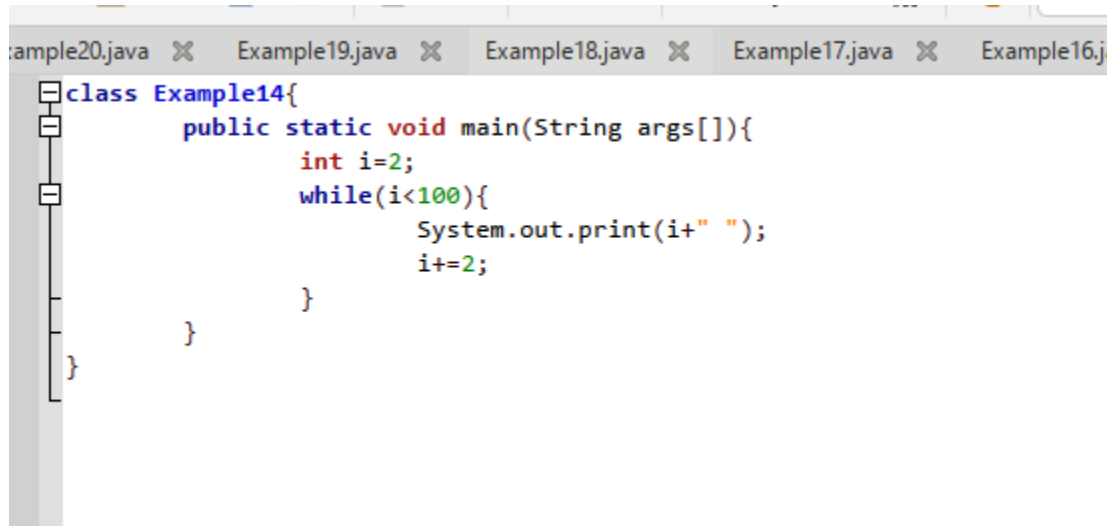
```
Example19.java  X  Example18.java  X  Example17.java  X  Example16.java  X
class Example13{
    public static void main(String args[]){
        int i =1;
        while(i<=100){
            System.out.print(i+" ");
            i++;
        }
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd  X
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43
44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83
84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

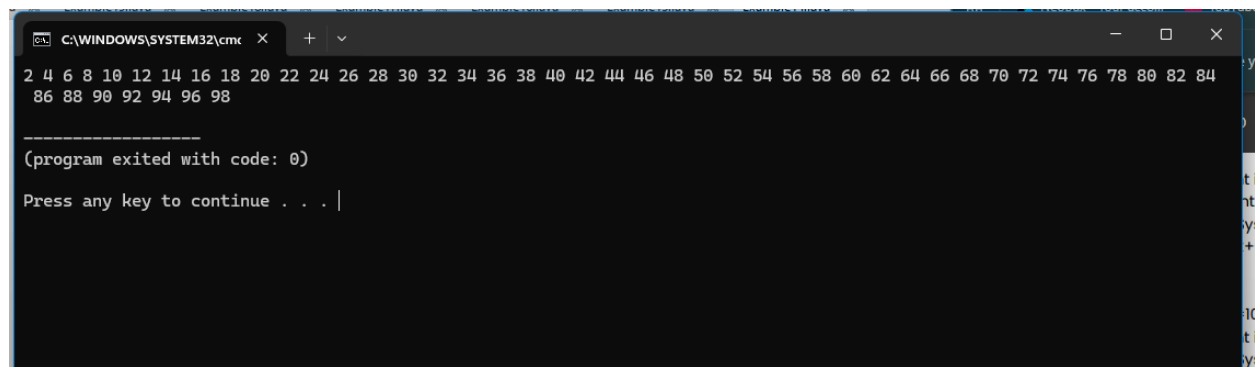
-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q14.



```
Example20.java X Example19.java X Example18.java X Example17.java X Example16.j
class Example14{
    public static void main(String args[]){
        int i=2;
        while(i<100){
            System.out.print(i+" ");
            i+=2;
        }
    }
}
```

Output.



```
C:\WINDOWS\SYSTEM32\cmd X + v
2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50 52 54 56 58 60 62 64 66 68 70 72 74 76 78 80 82 84
86 88 90 92 94 96 98

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q15.

Output.

Q16.

- 1 `public static void main(String[] args) {
 for (int i = 0; i < 10; i++){
 System.out.println(i);
 }
 }` output - 0 1 2 3 4 5 6 7 8 9
- 2 `for (int i = 0; i < 10; i++){
 System.out.println(i++);
 }` output - 0 2 4 6 8
- 3 `for (int i = 0; i < 10; i++){
 System.out.println(++i);
 }` output - 1 3 5 7 9
- 4 `int j=0;
while(j<10){
 System.out.println(j);
 j++;
}` output - 0 1 2 3 4 5 6 7 8 9
- 5 `int k=0;
while(k<10){
 System.out.println(k++);
}` output - 0 1 2 3 4 5 6 7 8 9
- 6 `int m=0;
while(m<10){
 System.out.println(++m); //6
}` output - 1 2 3 4 5 6 7 8 9 10
- 7 `int n=0;
while(n++<10){
 System.out.println(n); //7
}` output - 1 2 3 4 5 6 7 8 9 10

8

```
int p=0;
while(++p<10){
    System.out.println(p); //8
}
```

output - 1 2 3 4 5 6 7 8 9

9

```
int q=0;
do{
    System.out.println(q++); //9
}while(q<10);
```

output - 0 1 2 3 4 5 6 7 8 9

10

```
int s=0;
do{
    System.out.println(++s); //10
}while(s<10);
```

output - 1 2 3 4 5 6 7 8 9 10

11

```
int t=0;
do{
    System.out.println(t); //11
}while(t++<10);
```

output - 1 2 3 4 5 6 7 8 9 10

12

```
int u=0;
do{
    System.out.println(u); //12
}while(++u<10);
```

output - 1 2 3 4 5 6 7 8 9

Q17.

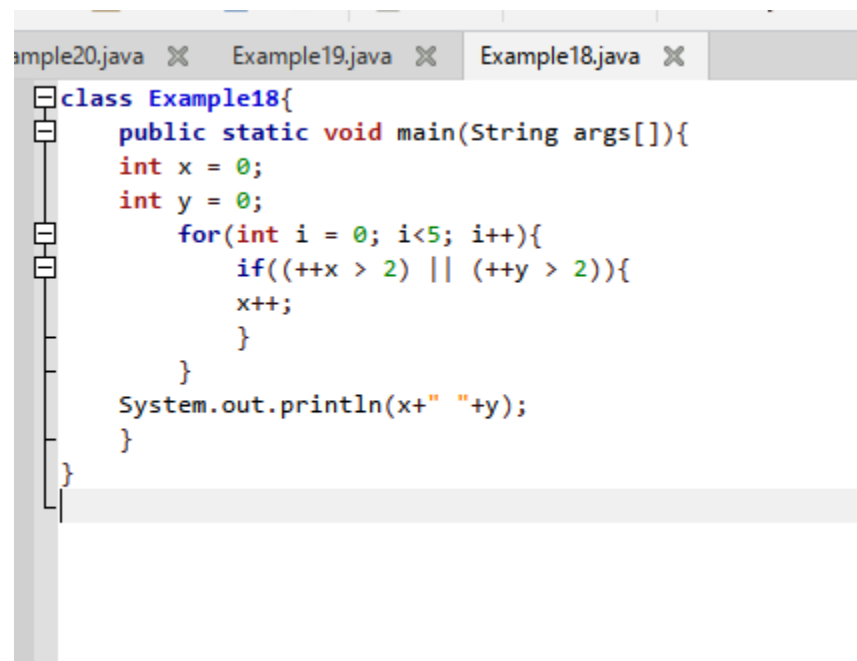
```
1 import java.util.*;
2 class Example17 {
3     public static void main(String args[]) {
4         Scanner input = new Scanner(System.in);
5         int n5k = 0, n2k = 0, n1k = 0, n500 = 0, n200 = 0, n100 = 0;
6         int n50 = 0, n20 = 0, n10 = 0, n5 = 0, n2 = 0, c1 = 0;
7         System.out.print("Input number of Employees: ");
8         int numEmp = input.nextInt();
9         for (int i = 1; i <= numEmp; i++) {
10             System.out.print("Input salary " + i + " : ");
11             int salary = input.nextInt();
12             n5k += salary / 5000;
13             salary %= 5000;
14             n2k += salary / 2000;
15             salary %= 2000;
16             n1k += salary / 1000;
17             salary %= 1000;
18             n500 += salary / 500;
19             salary %= 500;
20             n200 += salary / 200;
21             salary %= 200;
22             n100 += salary / 100;
23             salary %= 100;
24             n50 += salary / 50;
25             salary %= 50;
26             n20 += salary / 20;
27             salary %= 20;
28             n10 += salary / 10;
29             salary %= 10;
30             n5 += salary / 5;
31             salary %= 5;
32             n2 += salary / 2;
33             salary %= 2;
34             c1 += salary / 1;
35             salary %= 1;
36             System.out.println("R5000 notes : \t" + n5k);
37             System.out.println("R2000 notes : \t" + n2k);
38             System.out.println("R1000 notes : \t" + n1k);
39             System.out.println("R500 notes : \t" + n500);
40             System.out.println("R200 notes : \t" + n200);
41             System.out.println("R100 notes : \t" + n100);
42             System.out.println("R50 notes : \t" + n50);
43             System.out.println("R20 notes : \t" + n20);
44             System.out.println("R10 coins : \t" + n10);
45             System.out.println("R5 coins : \t" + n5);
46             System.out.println("R2 coins : \t" + n2);
47             System.out.println("R1 Coins : \t" + c1);
48             System.out.println();
49         }
50     }
51 }
```

```
C:\WINDOWS\SYSTEM32\cmd
Input number of Employees: 3
Input salary 1 : 345345
R5000 notes : 69
R2000 notes : 0
R1000 notes : 0
R500 notes : 0
R200 notes : 1
R100 notes : 1
R50 notes : 0
R20 notes : 2
R10 coins : 0
R5 coins : 1
R2 coins : 0
R1 Coins : 0

Input salary 2 : 345345
R5000 notes : 138
R2000 notes : 0
R1000 notes : 0
R500 notes : 0
R200 notes : 2
R100 notes : 2
R50 notes : 0
R20 notes : 4
R10 coins : 0
R5 coins : 2
R2 coins : 0
R1 Coins : 0

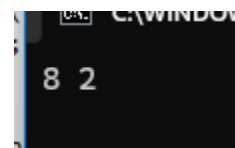
Input salary 3 : 34535
```

Q18.



```
class Example18{
    public static void main(String args[]){
        int x = 0;
        int y = 0;
        for(int i = 0; i<5; i++){
            if((++x > 2) || (++y > 2)){
                x++;
            }
        }
        System.out.println(x+ " "+y);
    }
}
```

Output



```
C:\WINDOW
8 2
```

Q19.

19. Which of the following code fragments are legal?

- A. `int i=0;`
`for(int i=0;i<10;i++){}`
- B. `for(int i=0;i<10;i++){}`
`for(int i=0;i<10;i++){}`
- C. `for(int i=0;i<10;i++){`
`int i=3;`
`}`
- D. `{int i=0;`
`for(int i=0;i<10;i++){}`

Output.

```
ample20.java x Example19.java x
class Example19{
    public static void main(String args[]){
        int i=0;
        for(int i=0;i<10;i++){ //illega

        //-----

        for(int i=0;i<10;i++){ //lega
        for(int i=0;i<10;i++){

        //-----

        for(int i=0;i<10;i++){ //illega
        int i=3;
        }

        //-----

        {int i=0;}
        for(int i=0;i<10;i++){ //lega

        }
    }
}
```

Q20.

20. Which of the following code fragment can be used to print integer numbers 100 to 109.

A.

```
for(int i=100;i<110;i++){  
    System.out.println(i);  
}
```

B.

```
for(int i=100;i<110;i++){  
    System.out.println(i++);  
}
```

C.

```
for(int i=0;i<10;i++){  
    int k=100;  
    System.out.println(i+k);  
}
```

D.

```
for(int i=0;i<10;i++){  
    int k=100;  
    System.out.println(k);  
    k++;  
}
```

E.

```
int k=100;  
for(int i=0;i<10;i++){  
    System.out.println(k);  
    k++;  
}
```

Answer – A , C, E.

Q21.

21. What is the output of the following code fragments?

- A.

```
for(int i=0;;i++){  
    System.out.println("i: "+i);  
}
```
- B.

```
int i=0;  
for(;i<10;){  
    System.out.println("i: "+i++);  
}
```
- C.

```
for(int i=0;i<10;i++){  
    System.out.println("i: "+i++);  
}
```
- D.

```
for(char ch='A';ch<9;ch++){  
    System.out.println("ch: "+ch);  
}
```
- E.

```
for(int i=0,j=10;i<=j;i++,j--){  
    System.out.println(i+" "+j);  
}
```
- F.

```
for(int i=0;i<128;i++){  
    System.out.println((char)i+" "+i);  
}
```
- G.

```
int x=1;  
for(int i=0;i<10;i++){  
    int k=100;  
    System.out.println(x+k);  
    k++;  
}
```

Answers –

A – infinite loop.

B -

```
i : 0  
i : 1  
i : 2  
i : 3  
i : 4  
i : 5  
i : 6  
i : 7  
i : 8  
i : 9
```

C -

```
i : 0  
i : 2  
i : 4  
i : 6  
i : 8
```

D -

```
ch : A
ch : B
ch : C
ch : D
ch : E
ch : F
ch : G
ch : H
ch : I
ch : J
ch : K
ch : L
ch : M
ch : N
ch : O
ch : P
ch : Q
ch : R
ch : S
ch : T
ch : U
ch : V
ch : W
ch : X
ch : Y
ch : Z
```

E -

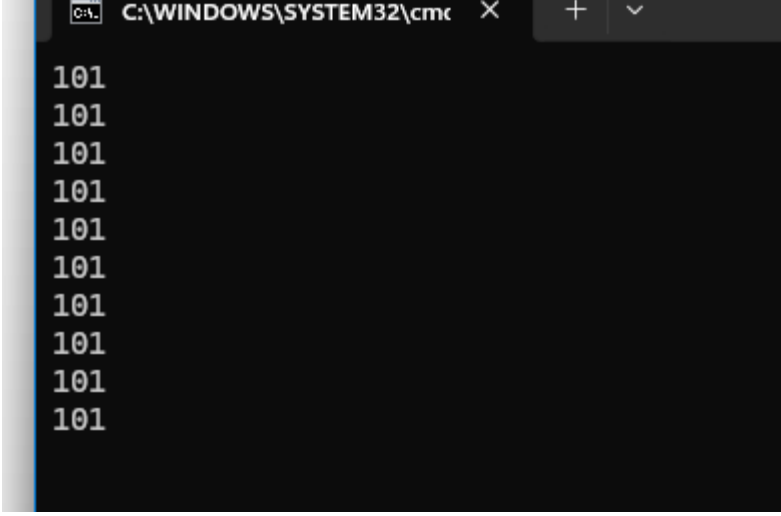
```
0 10
1 9
2 8
3 7
4 6
5 5
```

F -

```
C:\WINDOWS\SYSTEM32\cmd.exe
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
? 26
7
28
29
30
31
32
! 33
" 34
# 35
$ 36
% 37
& 38
' 39
( 40
) 41
* 42
+ 43
, 44
- 45
. 46
/ 47
```

Giving full ascii table

G -



A screenshot of a Windows command prompt window. The title bar at the top shows the path 'C:\WINDOWS\SYSTEM32\cmd' followed by a close button 'X' and window control buttons '+', '-', and 'v'. The command prompt itself has a black background with white text. It displays a series of ten '101' outputs, one per line, indicating a loop or a series of identical commands being executed.

```
C:\WINDOWS\SYSTEM32\cmd X + - v
101
101
101
101
101
101
101
101
101
101
```

Q22.

22. Which of the following loop declarations are legal?

- | | |
|---|---|
| A. <code>for(int i=0;i<10;i++){}</code> | B. <code>for(i=0;i<10;i++){}</code> |
| C. <code>for(int i=0;;i++){}</code> | D. <code>for(int i=0;i<10;){}</code> |
| E. <code>for(double d=0;d<10;d++){}</code> | F. <code>for(;;){}</code> |
| G. <code>for(byte b=0;b<10;b++){}</code> | H. <code>for(;int i=0;){}</code> |
| I. <code>for(int i=0,j=10;i++;j++){}</code> | J. <code>for(int i=0;;){}</code> |
| K. <code>for(char ch='A'; ch>92;ch++){}</code> | |

Answer – A, C, D, E, F, G, J, K.

Q23.

```
pie20.java  Example21.java  Example22.java  Example23.java
import java.util.*;
class Example23{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        int i = 1;
        System.out.print("Input an integer number "+i+", input use -1 to terminate : ");
        int num = input.nextInt();
        int g1k =0;
        int l1k =0;

        while(num!=-1){
            if (num>=1000){
                g1k++;
            }else{
                l1k++;
            }
            i++;
            System.out.print("Input an integer number "+i+", input use -1 to terminate : ");
            num = input.nextInt();
        }
        System.out.println(g1k+" numbers were less than 1000");
        System.out.println(l1k+" numbers were greater than 1000");
    }
}
```

Output.

```
Input an integer number 1, input use -1 to terminate : 2344
Input an integer number 2, input use -1 to terminate : 34
Input an integer number 3, input use -1 to terminate : 546
Input an integer number 4, input use -1 to terminate : 6567
Input an integer number 5, input use -1 to terminate : 676
Input an integer number 6, input use -1 to terminate : 34
Input an integer number 7, input use -1 to terminate : -1
2 numbers were less than 1000
4 numbers were greater than 1000
```

```
-----
(program exited with code: 0)
```

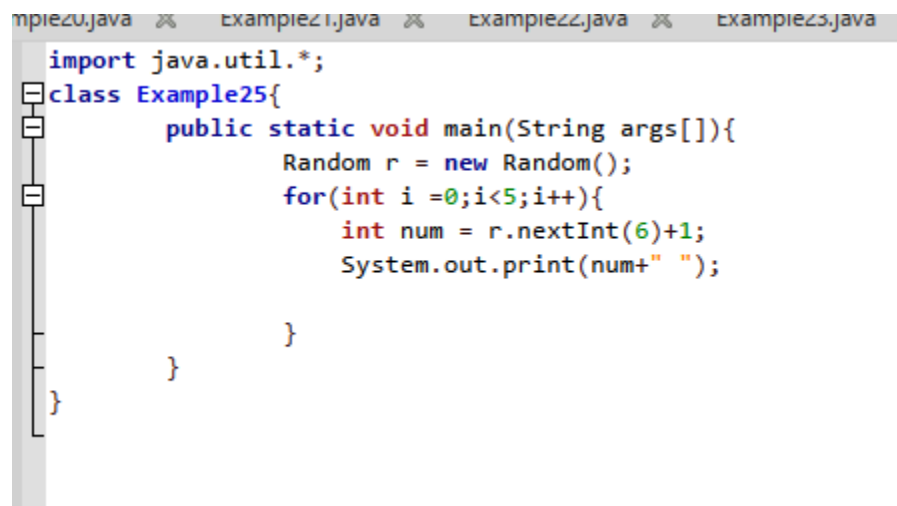
Q24.

```
Example20.java ✕ Example21.java ✕ Example22.java ✕ Example23.java ✕ Example24.java ✕
1  import java.util.*;
2  class Example24{
3      public static void main(String args[]){
4          Random ran = new Random();
5          int head = 0;
6          int tail = 0;
7          for(int i = 0;i<10;i++){
8              boolean tOF = ran.nextBoolean();
9              if(tOF){
10                 System.out.println("HEAD");
11                 tail++;
12             }else{
13                 System.out.println("TAIL");
14                 head++;
15             }
16         }
17         System.out.print("Statistics: "+head+" Heads and "+tail+" Tails");
18     }
19 }
20 }
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd ✕ + v
HEAD
HEAD
TAIL
TAIL
HEAD
HEAD
HEAD
TAIL
TAIL
HEAD
Statistics: 4 Heads and 6 Tails
```

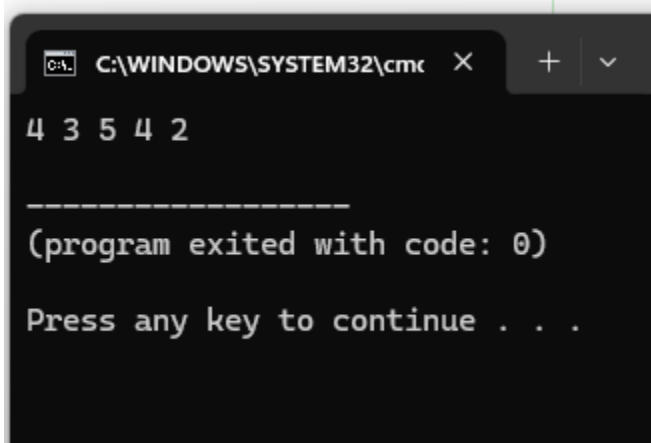
Q25.



```
import java.util.*;

class Example25{
    public static void main(String args[]){
        Random r = new Random();
        for(int i =0;i<5;i++){
            int num = r.nextInt(6)+1;
            System.out.print(num+" ");
        }
    }
}
```

Output.



```
C:\WINDOWS\SYSTEM32\cmd

4 3 5 4 2

-----
(program exited with code: 0)
Press any key to continue . . .
```

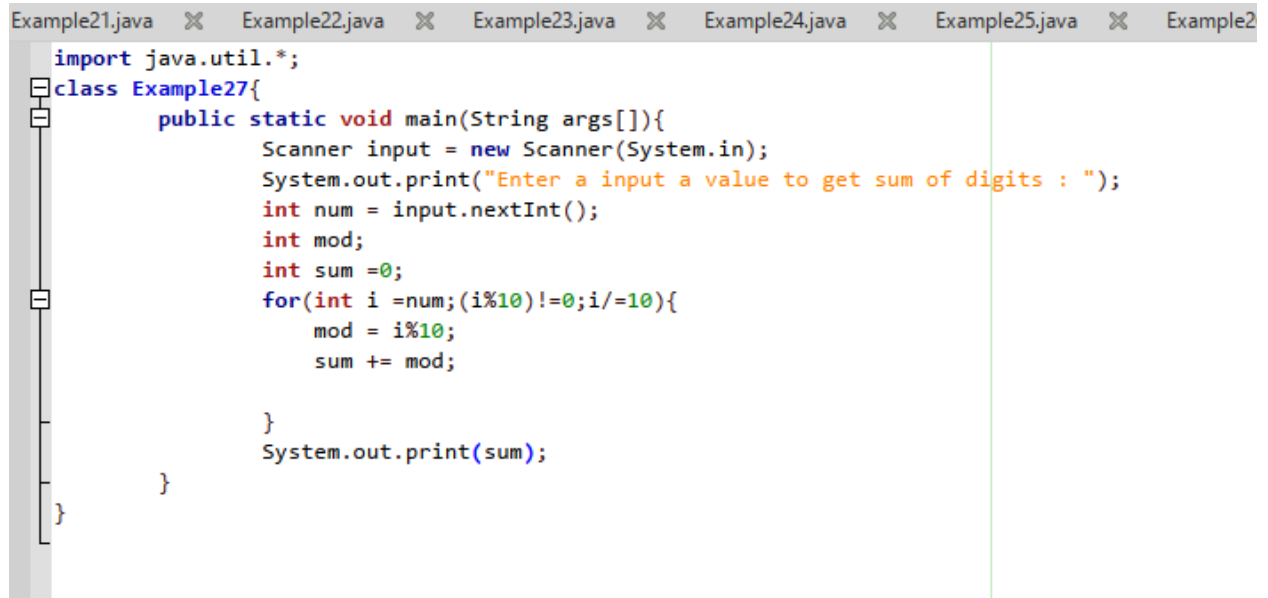
Q26.

```
Example20.java ✕ Example21.java ✕ Example22.java ✕ Example23.java ✕ Example24.java ✕ Example25.java ✕  
import java.util.*;  
class Example26{  
    public static void main(String args[]){  
        Random ran = new Random();  
        Scanner input = new Scanner(System.in);  
        System.out.print("Enter number that you want to get from dice : ");  
        int uI = input.nextInt();  
        int rDN = ran.nextInt(6)+1;  
        while(rDN!=uI){  
            System.out.println(rDN);  
            rDN = ran.nextInt(6)+1;  
        }  
    }  
}
```

Output.

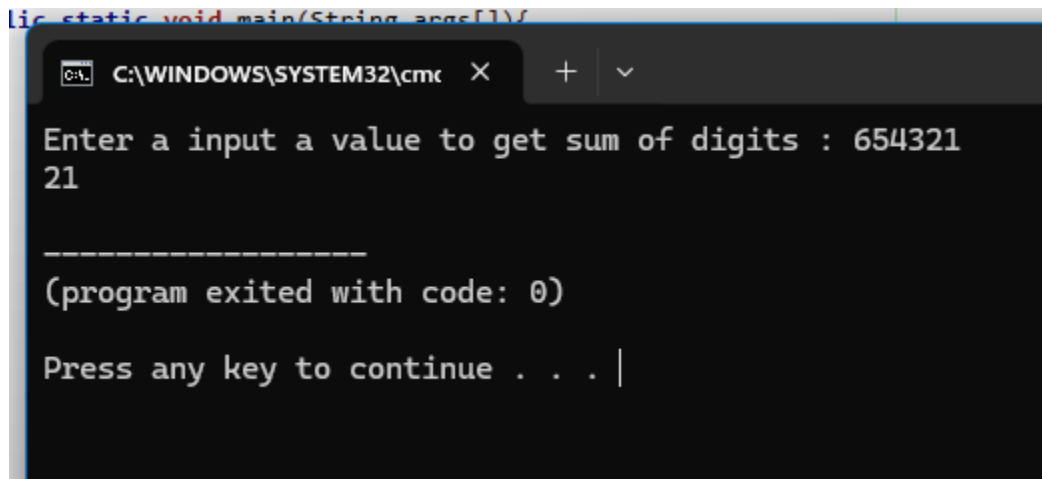
```
C:\WINDOWS\SYSTEM32\cmd.exe ✕ + v  
Enter number that you want to get from dice : 4  
6  
6  
2  
6  
6  
5  
1  
5  
  
-----  
(program exited with code: 0)  
Press any key to continue . . .
```

Q27.



```
import java.util.*;
class Example27{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter a input a value to get sum of digits : ");
        int num = input.nextInt();
        int mod;
        int sum =0;
        for(int i =num;(i%10)!=0;i/=10){
            mod = i%10;
            sum += mod;
        }
        System.out.print(sum);
    }
}
```

Output.



```
C:\WINDOWS\SYSTEM32\cmd X + v
Enter a input a value to get sum of digits : 654321
21
-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q28.

```
xample22.java  Example23.java  Example24.java  Example25.java  Ex
import java.util.*;
class Example28{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter an integer number : ");
        int num = input.nextInt();

        int rNum = 0;
        for(int i = num; i%10!=0; i/=10){
            int mod = i%10;
            rNum = (rNum*10)+mod;
        }

        System.out.print(rNum);
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd  +  v
Enter an integer number : 1234
4321

-----
t (program exited with code: 0)

Press any key to continue . . . |
```

Q29.

```
xample29.java x Example30.java x Example31.java x Example32.java x Example33.java x Example34.java x Example35.java x
import java.util.*;
class Example29 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("How many bottles of beer are on the wall: ");
        int tBottles = input.nextInt();
        System.out.println();
        for (int i = tBottles; i >= 1; i--) {
            if (i > 1) {
                System.out.println(i + " bottles of beer on the wall, " + i + " bottles of beer.");
            } else {
                System.out.println("1 bottle of beer on the wall, 1 bottle of beer.");
            }

            if (i - 1 > 1) {
                System.out.println("Take one down, pass it around, " + (i - 1) + " bottles of beer on the wall.");
            } else if (i - 1 == 1) {
                System.out.println("Take one down, pass it around, 1 bottle of beer on the wall.");
            } else {
                System.out.println("Take one down, pass it around, no more bottles of beer on the wall.");
            }

            System.out.println();
        }
    }
}
```

Output

```
C:\WINDOWS\SYSTEM32\cmd.exe
Q:
xam
4 bottles of beer on the wall, 4 bottles of beer.
Take one down, pass it around, 3 bottles of beer on the wall.
3 bottles of beer on the wall, 3 bottles of beer.
Take one down, pass it around, 2 bottles of beer on the wall.
2 bottles of beer on the wall, 2 bottles of beer.
Take one down, pass it around, 1 bottle of beer on the wall.
1 bottle of beer on the wall, 1 bottle of beer.
Take one down, pass it around, no more bottles of beer on the wall.

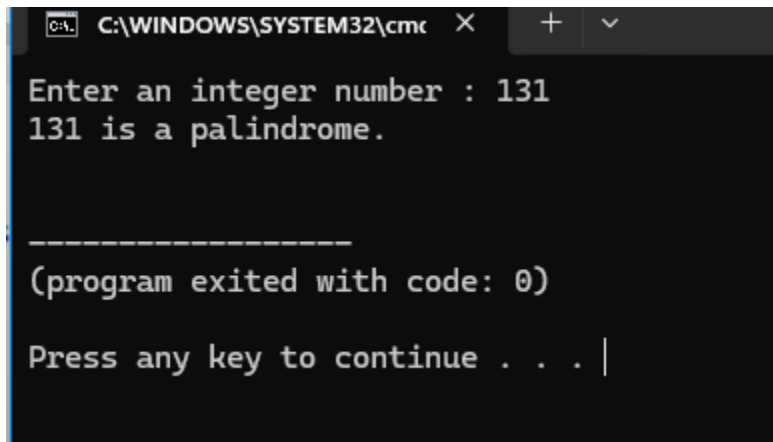
-----
(program exited with code: 0)
```

Q30.

```
import java.util.*;

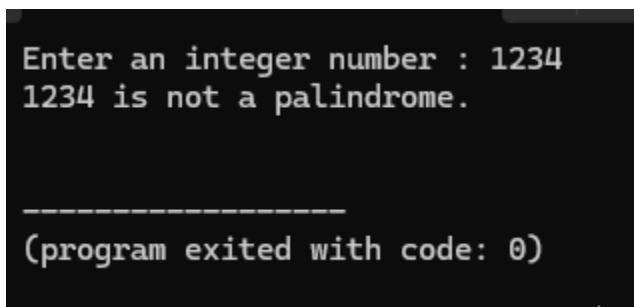
class Example30 {
    public static void main(String args[]) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter an integer number : ");
        int num = input.nextInt();
        int rNum = 0;
        for(int i = num; i != 0; i /= 10){
            int mod = i % 10;
            rNum = (rNum * 10) + mod;
        }
        if (num == rNum) {
            System.out.println(num + " is a palindrome.");
        } else {
            System.out.println(num + " is not a palindrome.");
        }
    }
}
```

Output.



```
C:\WINDOWS\SYSTEM32\cmd.exe
Enter an integer number : 131
131 is a palindrome.

-----
(program exited with code: 0)
Press any key to continue . . . |
```



```
Enter an integer number : 1234
1234 is not a palindrome.

-----
(program exited with code: 0)
```

Q31.

```
Example29.java  Example30.java  Example31.java  Example32.java  Example33.java  Example34.java
import java.util.*;
class Example31 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("How many classes : ");
        int no_cla = input.nextInt();
        for(int i = 1; i<=no_cla ; i++) {
            System.out.println("\n--- Class " + i + " ---");
            System.out.print("No of Students : ");
            int num = input.nextInt();
            if(num == -1) {
                System.out.println("Program terminated.");
                break;
            }
            double tot= 0;
            double max= 0;
            double min = 0;
            for(int x=1; x<=num; x++) {
                System.out.print("Enter " + x + " marks : ");
                double marks = input.nextDouble();
                tot += marks;
                if(x == 1) {
                    max = marks;
                    min = marks;
                } else {
                    if(marks > max) {
                        max = marks;
                    }
                    if(marks < min) {
                        min = marks;
                    }
                }
            }
            double average = total / num;
            System.out.println("\nNo of Students : " + num);
            System.out.println("Total marks : " + (int)tot);
            System.out.println("Maximum : " + (int)max);
            System.out.println("Minimum : " + (int)min);
            System.out.printf("Average : %.2f\n", average);
        }
    }
}
```

Output.

How many classes : 4

--- Class 1 ---

No of Students : 3

Enter 1 marks : 23

Enter 2 marks : 45

Enter 3 marks : 78

No of Students : 3

Total marks : 146

Maximum : 78

Minimum : 23

Average : 48.67

--- Class 2 ---

No of Students : 1

Enter 1 marks : 24

No of Students : 1

Total marks : 24

Maximum : 24

Minimum : 24

Average : 24.00

--- Class 3 ---

No of Students : 2

Enter 1 marks : 34

Enter 2 marks : 87

No of Students : 2

Total marks : 121

Maximum : 87

Minimum : 34

Average : 60.50

--- Class 4 ---

No of Students : 5

Enter 1 marks : 23

Enter 2 marks : 76

Enter 3 marks : 98

Enter 4 marks : 45

Enter 5 marks : 67

No of Students : 5

Total marks : 309

Maximum : 98

Minimum : 23

Average : 61.80

ment Project

e30.java

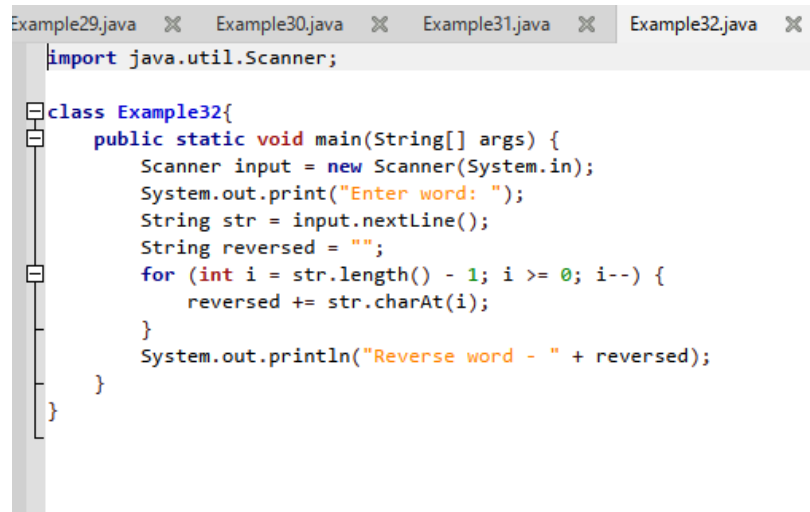
```
void main(S
ut = new Sc
print("How
+ input.nex
1; i<=no_c
out.println
out.println
= input.ne
-1) {
em.out.pri
k;
```

```
tot= 0;
max= 0;
min = 0;
x=1; x<=nu
em.out.pri
le marks =
+= marks;
<= 1) {
max = mark
min = mark
lse {
if(marks >
max =
}
if(marks <
min =
}
```

```
average = t
out.println
out.println
out.println
out.println
out.printf(
```

va" (in dir
d successfu

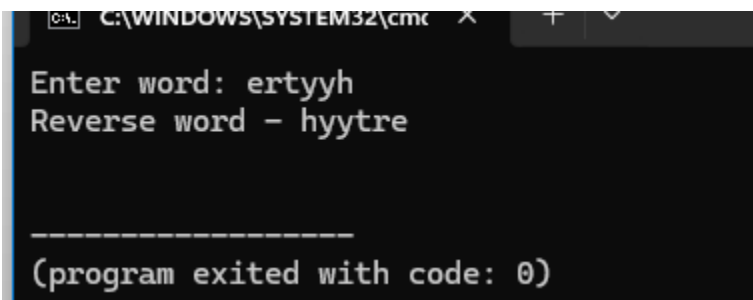
Q32.



```
import java.util.Scanner;

class Example32{
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter word: ");
        String str = input.nextLine();
        String reversed = "";
        for (int i = str.length() - 1; i >= 0; i--) {
            reversed += str.charAt(i);
        }
        System.out.println("Reverse word - " + reversed);
    }
}
```

Output.



```
C:\WINDOWS\SYSTEM32\cmd.exe
Enter word: ertyyh
Reverse word - hyytre

-----
(program exited with code: 0)
```

Q33.

```
Example29.java  Example30.java  Example31.java  Example32.java  Example33.java
import java.util.*;
class Example33{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter a string: ");
        String start = input.next();
        String reversed = "";
        for(int i = start.length() - 1; i >= 0; i--){
            reversed += start.charAt(i);
        }
        if(start.equals(reversed)){
            System.out.println(start + " is a Palindrome.");
        }else{
            System.out.println(start + " is NOT a Palindrome.");
        }
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd  +  v
Enter a string: uyhyuguy
uyhyuguy is NOT a Palindrome.

-----
in) (program exited with code: 0)
);
Press any key to continue . . . |
```

```
C:\WINDOWS\SYSTEM32\cmd  +  v
Enter a string: wew
wew is a Palindrome.

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q34.

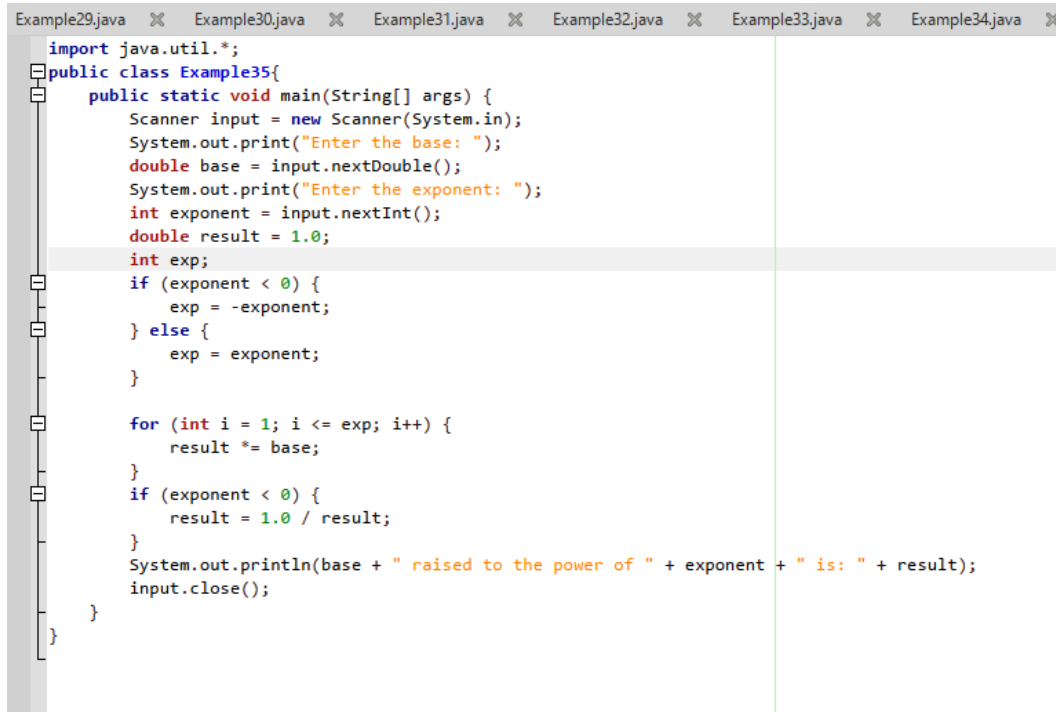
```
import java.util.*;
class Example34{
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter a string: ");
        String str = input.nextLine();
        System.out.print("Enter a character to search: ");
        char targetChar = input.next().charAt(0);
        int count = 0;
        for (int i = 0; i < str.length(); i++) {
            if (str.charAt(i) == targetChar) {
                count++;
            }
        }
        System.out.println("The character '" + targetChar + "' appears " + count + " time(s).");
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd.exe
Enter a string: wersrerereeregdfhfgherfgdre
Enter a character to search: e
The character 'e' appears 8 time(s).

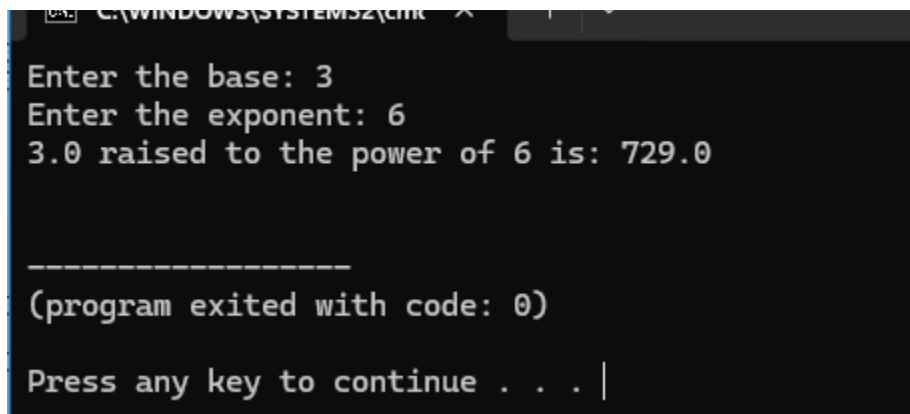
-----
(program exited with code: 0)
```

Q35.



```
import java.util.*;
public class Example35{
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the base: ");
        double base = input.nextDouble();
        System.out.print("Enter the exponent: ");
        int exponent = input.nextInt();
        double result = 1.0;
        int exp;
        if (exponent < 0) {
            exp = -exponent;
        } else {
            exp = exponent;
        }
        for (int i = 1; i <= exp; i++) {
            result *= base;
        }
        if (exponent < 0) {
            result = 1.0 / result;
        }
        System.out.println(base + " raised to the power of " + exponent + " is: " + result);
        input.close();
    }
}
```

Output.



```
C:\WINDOWS\SYSTEM32\cmd.exe
Enter the base: 3
Enter the exponent: 6
3.0 raised to the power of 6 is: 729.0

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q36.

```
import java.util.Scanner;
class Example36{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter a string: ");
        String str = input.nextLine();
        int vowels = 0;
        int consonants = 0;
        str = str.toLowerCase();
        for(int i = 0; i < str.length(); i++){
            char ch = str.charAt(i);
            if(ch >= 'a' && ch <= 'z'){
                if(ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u'){
                    vowels++;
                }else{
                    consonants++;
                }
            }
        }
        System.out.println("Number of Vowels: " + vowels);
        System.out.println("Number of Consonants: " + consonants);
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd  X  +  v
Enter a string: dfgdshttjhjbhvsnjssxbkolooop
Number of Vowels: 4
Number of Consonants: 26

-----
(program exited with code: 0)
```

Q37.

```

import java.util.Scanner;

class Example37{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the first positive integer: ");
        int num1 = input.nextInt();
        System.out.print("Enter the second positive integer: ");
        int num2 = input.nextInt();
        int gcd = 1;
        for(int i = 1; i <= num1 && i <= num2; i++){
            if(num1 % i == 0 && num2 % i == 0){
                gcd = i;
            }
        }
        System.out.println("The Greatest Common Divisor (GCD) of " + num1 + " and " + num2 + " is: " + gcd);
        input.close();
    }
}

```

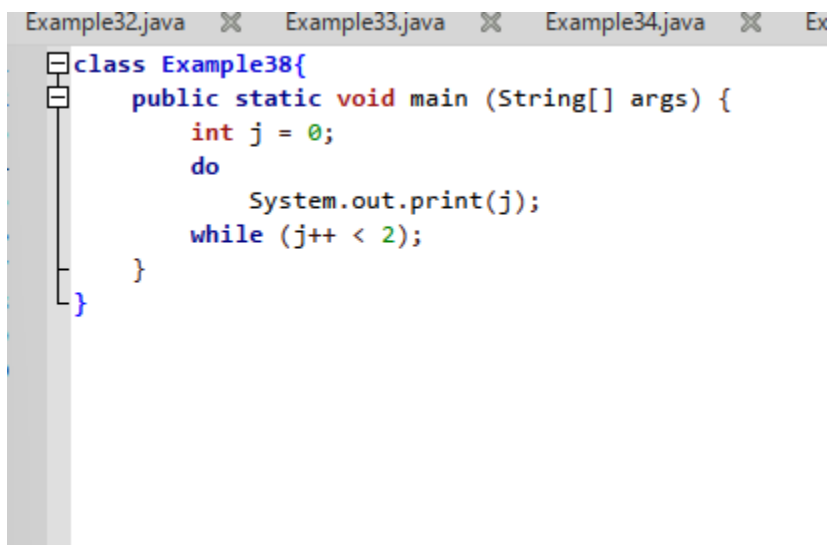
Output.

```
C:\WINDOWS\SYSTEM32\cmd.exe
Enter the first positive integer: 25
Enter the second positive integer: 67
The Greatest Common Divisor (GCD) of 25 and 67 is: 1

-----
(program exited with code: 0)

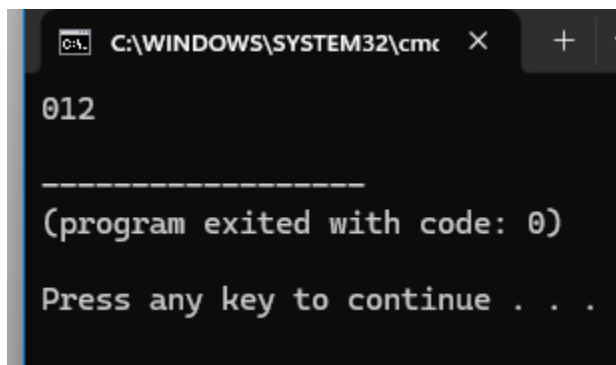
Press any key to continue . . .
```

Q38.



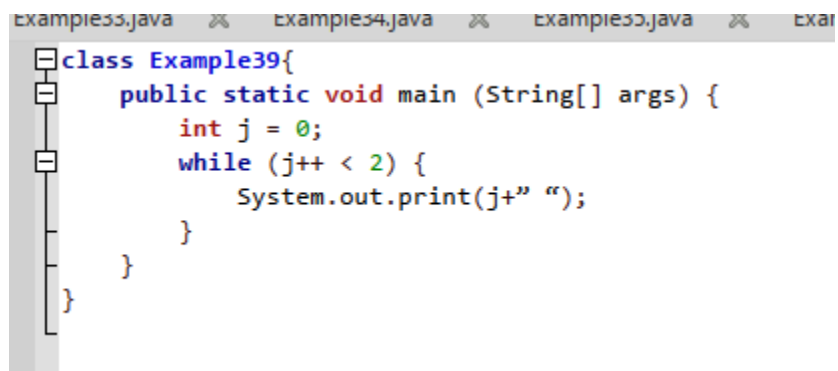
```
Example32.java Example33.java Example34.java Ex
class Example38{
    public static void main (String[] args) {
        int j = 0;
        do
            System.out.print(j);
        while (j++ < 2);
    }
}
```

Answer - B



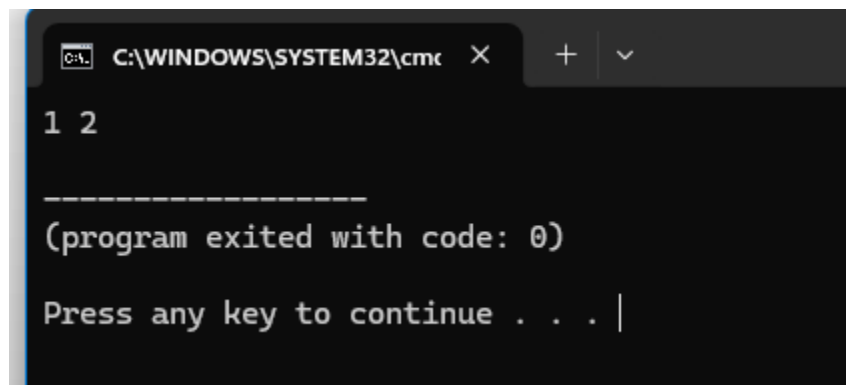
```
C:\WINDOWS\SYSTEM32\cmd
012
-----
(program exited with code: 0)
Press any key to continue . . .
```

Q39.



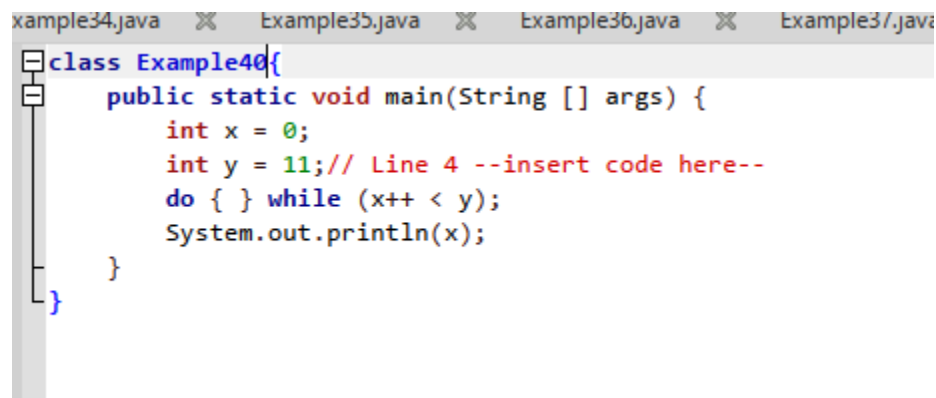
```
class Example39{
    public static void main (String[] args) {
        int j = 0;
        while (j++ < 2) {
            System.out.print(j+" ");
        }
    }
}
```

Output.



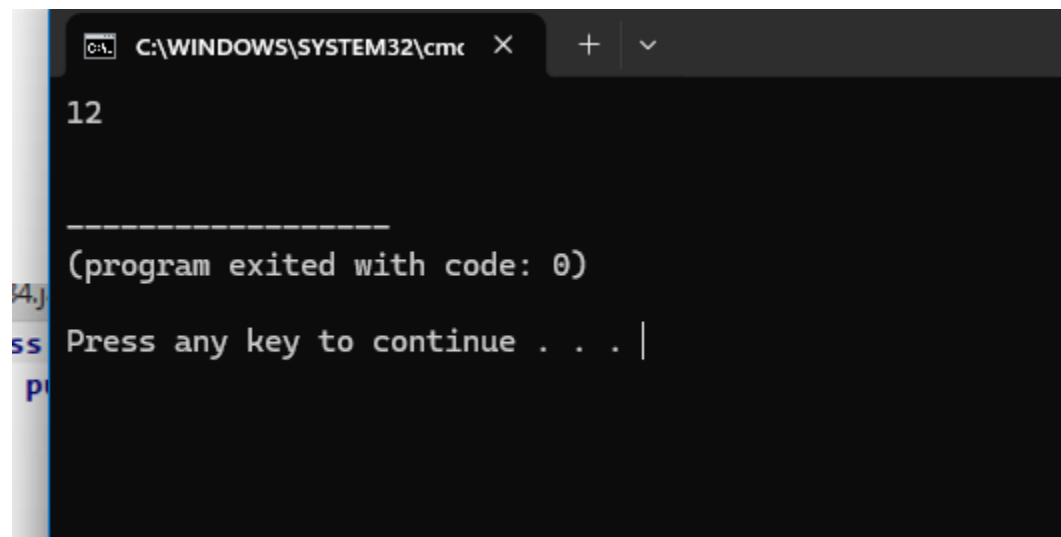
```
C:\WINDOWS\SYSTEM32\cmd.exe
1 2
-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q40.



```
class Example40{
    public static void main(String [] args) {
        int x = 0;
        int y = 11; // Line 4 --insert code here--
        do { } while (x++ < y);
        System.out.println(x);
    }
}
```

Output.



```
C:\WINDOWS\SYSTEM32\cmd.exe
12
-----
(program exited with code: 0)
Press any key to continue . . .
```

Q41.

```
Example35.java  Example36.java  Example37.java  Example38.java  Example39.java  Example40.java  X
class Example41{
    public static void main(String[] args) {
        for (int i = 0; i <= 4; i++) {
            for (int j = 0; j <= 4; j++) {
                System.out.println(i + " " + j + "\t" + j + " " + i + "\t" + (5 - i) + " " + j + "\t" + (5 - i) + " " + (5 - j));
            }
        }
    }
}
```

Output.

```
Example39.java  Example40.java  X  2  public
3  3  for
C:\WINDOWS\SYSTEM32\cmd  X  +  v
0 1 1 0 5 1 5 4
0 2 2 0 5 2 5 3
0 3 3 0 5 3 5 2
0 4 4 0 5 4 5 1
1 0 0 1 4 0 4 5
1 1 1 1 4 1 4 4
1 2 2 1 4 2 4 3
1 3 3 1 4 3 4 2
1 4 4 1 4 4 4 1
2 0 0 2 3 0 3 5
2 1 1 2 3 1 3 4
2 2 2 2 3 2 3 3
2 3 3 2 3 3 3 2
2 4 4 2 3 4 3 1
3 0 0 3 2 0 2 5
3 1 1 3 2 1 2 4
3 2 2 3 2 2 2 3
3 3 3 3 2 3 2 2
3 4 4 3 2 4 2 1
4 0 0 4 1 0 1 5
4 1 1 4 1 1 1 4
4 2 2 4 1 2 1 3
4 3 3 4 1 3 1 2
4 4 4 4 1 4 1 1

-----
(program exited with code: 0)
```

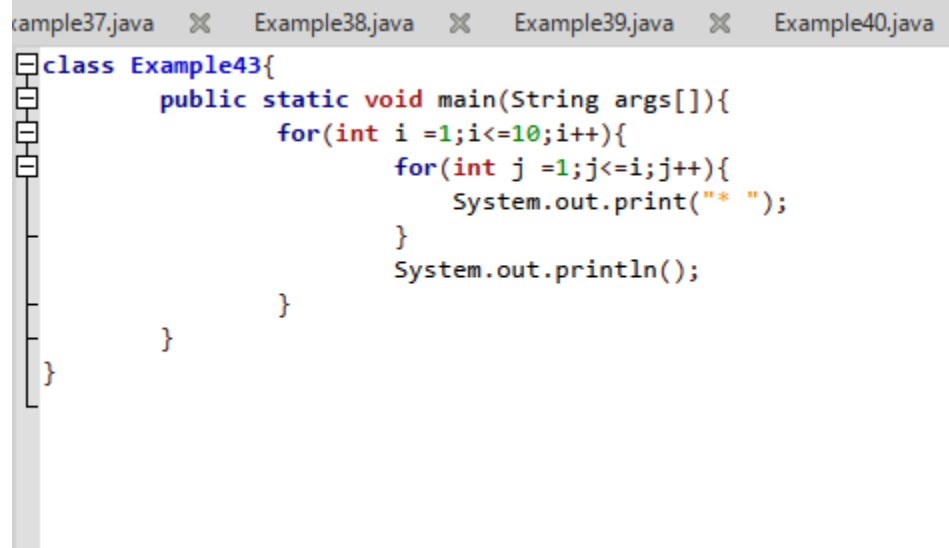
Q42.

```
class Example42{
    public static void main(String args[]){
        for(int i =0;i<10;i++){
            for(int j=0;j<10;j++){
                System.out.print("* ");
            }
            System.out.println();
            System.out.println();
        }
    }
}
```

Output.

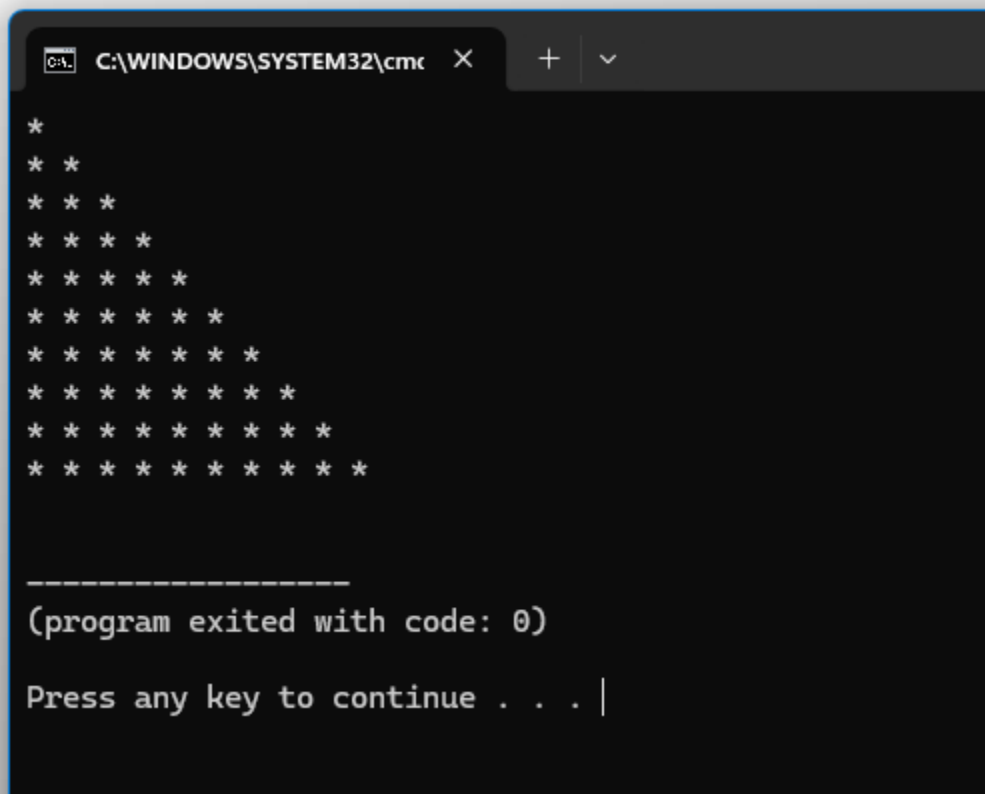
[illegible]

Q43.



```
class Example43{
    public static void main(String args[]){
        for(int i =1;i<=10;i++){
            for(int j =1;j<=i;j++){
                System.out.print("* ");
            }
            System.out.println();
        }
    }
}
```

Output.



```
*
* *
* * *
* * * *
* * * * *
* * * * * *
* * * * * * *
* * * * * * *
* * * * * * *
* * * * * * *
* * * * * * *
```

(program exited with code: 0)

Press any key to continue . . . |

Q44.

```
Example38.java  X  Example39.java  X  Example40.java  X  Example41.java  X
class Example44{
    public static void main(String args[]){
        for(int i =10;i>=1;i--){
            for(int j =i;j>=1;j--){
                System.out.print("* ");
            }
            System.out.println();
        }
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd  X  +  v
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * *
* * * *
* * *
* *
*
-----
(program exited with code: 0)
```

Q45.

```
ample51.java  Example50.java  Example49.java  Example48.java
import java.util.*;
class Example45 {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter the size : ");
        int size = input.nextInt();
        System.out.print(" * | ");
        for(int i = 1; i<=size;i++){
            System.out.print(" "+i+" ");
        }

        System.out.println();
        for(int b = 1; b<=size;b++){
            System.out.print("----");
        }
        System.out.println();
        for(int d = 1 ;d<=size;d++){
            System.out.print(" "+d+" | ");
            for(int j = 1 ;j<=size ; j++){
                System.out.print((d*j) + " ");
            }
            System.out.println();
        }
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd  +  v
Enter the size : 6
 * | 1 2 3 4 5 6
----
1 | 1 2 3 4 5 6
2 | 2 4 6 8 10 12
3 | 3 6 9 12 15 18
4 | 4 8 12 16 20 24
5 | 5 10 15 20 25 30
6 | 6 12 18 24 30 36

-----
(program exited with code: 0)
Press any key to continue . . .
```

Q46.

```
import java.util.*;
class Example46 {
    public static void main(String[] args){
        Scanner input =new Scanner(System.in);
        Random r = new Random();
        int ran_num = r.nextInt(100)+1;
        int num = -1;
        while (num != ran_num) {
            System.out.print("Enter your guessing number : ");
            num = input.nextInt();

            if(num > ran_num){
                System.out.println("Too high, try again.");
            } else if(num < ran_num){
                System.out.println("Too low, try again.");
            } else {
                System.out.println("Your Guessing is correct.");
            }
        }
    }
}
```

Output.

```
Enter your guessing number : 55
Too high, try again.
Enter your guessing number : 30
Too low, try again.
Enter your guessing number : 40
Too high, try again.
Enter your guessing number : 45
Too high, try again.
Enter your guessing number : 42
Too high, try again.
Enter your guessing number : 41
Too high, try again.
Enter your guessing number : 30
Too low, try again.
Enter your guessing number : 39
Too high, try again.
Enter your guessing number : 38
Too high, try again.
Enter your guessing number : 37
Too high, try again.
Enter your guessing number : 36
Too high, try again.
Enter your guessing number : 35
Your Guessing is correct.
-----
```

Q47.

```
Example51.java  Example50.java  Example49.java  Exa
class Example47 {
    public static void main(String[] args){
        for(int i = 0 ;i<=7;i++){
            for(int j = 1;j<=i;j++){
                System.out.print(j);
            }
            System.out.println();
        }
    }
}
```

Output.

```
1
12
123
1234
12345
123456
1234567
```

```
-----
(program exited with code: 0)
```

Q48.

```
class Example48 {  
    public static void main(String[] args) {  
        int i = 2;  
        while (i <= 100) {  
            int j = 2;  
            boolean prime = true;  
            while (j <= i / 2) {  
                if (i % j == 0) {  
                    prime = false;  
                    break;  
                }  
                j++;  
            }  
            if (prime) {  
                System.out.println(i);  
            }  
            i++;  
        }  
    }  
}
```

Output.

```
2  
3  
5  
7  
11  
13  
17  
19  
23  
29  
31  
37  
41  
43  
47  
53  
59  
61  
67  
71  
73  
79  
83  
89  
97
```

Q49.

```
import java.util.*;
class Example49{
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int num = input.nextInt();
        int ori_num = num;
        int sum = 0;
        while (num > 0) {
            int digit = num % 10;
            sum += (digit * digit * digit);
            num /= 10;
        }
        if (sum == ori_num) {
            System.out.println(ori_num + " is an Armstrong number.");
        } else {
            System.out.println(ori_num + " is not an Armstrong number.");
        }
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd
Enter a number: 345
345 is not an Armstrong number.

-----
(program exited with code: 0)
```

```
C:\WINDOWS\SYSTEM32\cmd
Enter a number: 371
371 is an Armstrong number.

-----
(program exited with code: 0)

Press any key to continue . . . |
```

Q50.

```
Example50.java Example50.java Example49.java Example40.java Example47.java Example40.java
class Example50 {
    public static void main(String[] args){
        int sum = 0 ;
        for(int i = 1;i<1000;i++){
            if(i %3==0 || i % 5==0){
                sum+=i;
            }
        }
        System.out.println(sum +" of numbers are multiples of 3 or 5 below 1000.");
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd.exe X + v
233168 of numbers are multiples of 3 or 5 below 1000.

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q51.

```
Example51.java  Example50.java  Example49.java  Example48.java  Example47.java  Example46.java  Example45.java
class Example51 {
    public static void main(String[] args) {
        long number = 1;
        boolean found = false;
        while (!found) {
            boolean isDivid = true;
            for (int i = 1; i <= 20; i++) {
                if (number % i != 0) {
                    isDivid = false;
                    break;
                }
            }
            if (isDivid) {
                System.out.println("the smallest positive number that is evenly divisible by all of the numbers from 1 to 20 is: " + number);
                found = true;
            } else {
                number++;
            }
        }
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd  +
the smallest positive number that is evenly divisible by all of the numbers from 1 to 20 is: 232792560

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q52.

```
ample55.java  X Example54.java  X Example53.java  X Example52.java  X Example51.java  X
import java.util.*;
class Example52 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter words: ");
        String input_num = input.nextLine();

        int count = 0;
        int i = 0;
        while (i < input_num.length()) {
            char ch = input_num.charAt(i);
            if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u' ||
                ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch == 'U') {
                count++;
            }
            i++;
        }
        System.out.println("Number of vowels: " + count);
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd  X + v
Enter words: dfbgbfnmhuteshsrh
Number of vowels: 2

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q53.

```
import java.util.*;
class Example53 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter base: ");
        int base = input.nextInt();

        System.out.print("Enter exponent: ");
        int exponent = input.nextInt();

        long result = 1;
        int i = 0;

        while(i < exponent){
            result *= base;
            i++;
        }
        System.out.println("Result: " + result);
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd.exe
Enter base: 4
Enter exponent: 6
Result: 4096

-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q54.

```
Example55.java  Example54.java  Example53.java  Exa
class Example54 {
    public static void main(String[] args) {
        for (int i=1; i<=100;i++) {
            int count = 0;

            for(int j=1;j<=i;j++){
                if (i%j==0) {
                    count++;
                }
            }
            if(count==2){
                System.out.println(i);
            }
        }
    }
}
```

Output.

```
2
3
5
7
11
13
17
19
23
29
31
37
41
43
47
53
59
61
67
71
73
79
83
89
97
```

Q55.

```
class Example55 {
    public static void main(String[] args){
        System.out.println("Starting random point generation inside the unit circle:\n");

        int totPoints = 0;

        while(true){
            double x = Math.random()*2.0 - 1.0;
            if(x < -1.0){
                x = -1.0;
            }else if (x > 1.0){
                x = 1.0;
            }

            double y = Math.random() * 2.0 - 1.0;
            if(y<-1.0){
                y=-1.0;
            }else if(y > 1.0){
                y = 1.0;
            }
            totPoints++;
            if(x * x + y * y <= 1.0){
                System.out.printf("Point %d (Inside): x = %.1f, y = %.1f\n",totPoints, x, y);
            }else{
                System.out.printf("Caught an outsider! x = %.1f, y = %.1f (Outside the circle)\n", x, y);
                System.out.println(" Total generated points : " + totPoints);
                break;
            }
        }
    }
}
```

Output.

```
Starting random point generation inside the unit circle:

Point 1 (Inside): x = -0.1, y = 0.4
Caught an outsider! x = 0.9, y = 0.9 (Outside the circle)
Total generated points : 2

-----
(program exited with code: 0)
Press any key to continue . . . |
```

```
Starting random point generation inside the unit circle:

Point 1 (Inside): x = -0.1, y = 0.5
Point 2 (Inside): x = -0.3, y = 0.2
Point 3 (Inside): x = -0.9, y = 0.2
Point 4 (Inside): x = 0.5, y = 0.7
Point 5 (Inside): x = -0.6, y = -0.3
Point 6 (Inside): x = -0.5, y = 0.9
Point 7 (Inside): x = 0.1, y = 0.2
Point 8 (Inside): x = -0.5, y = 0.4
Point 9 (Inside): x = -0.2, y = 0.7
Point 10 (Inside): x = 0.4, y = -0.7
Point 11 (Inside): x = -0.5, y = 0.4
Point 12 (Inside): x = -0.2, y = -0.6
Caught an outsider! x = 1.0, y = -0.2 (Outside the circle)
Total generated points : 13
```

Q56.

56. What will be the output when you compile and run the following program?

Explain your answer.

```
class Example{
    public static void main(String[] args) {
        int a,b,c,d,e;
        for (int i = 0; i < 10; i++){
            a=i;
        }

        int j=0;
        while(j++<10){
            b=j;
        }

        int k=0;
        do{
            c=j;
        }while(j++<10);

        int m=0;
        switch(m){
            case 0 : d=0;
        }

        int n=0;
        sw  itch(n){
            default : e=0;
        }

        System.out.println(a); //Line 24
        System.out.println(b); //Line 25
        System.out.println(c); //Line 26
        System.out.println(d); //Line 27
        System.out.println(e); //Line 28
    }
}
```

Explain - syntax error ,because of type error in “sw itch” it has to be Switch.

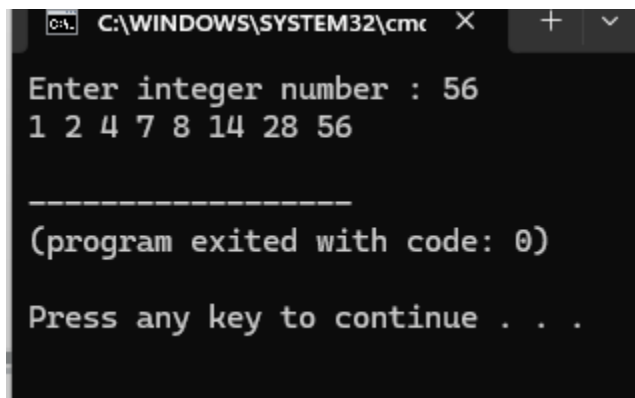
```
35 }
36 }

Status javac "Example56.java" (in directory: D:\ICD124\DAY 5\N\Assignment)
Compiler Example56.java:25: error: ';' expected
           sw  itch(n){
               ^
Messages Example56.java:25: error: not a statement
           sw  itch(n){
               ^
Cribble Example56.java:25: error: ';' expected
           sw  itch(n){
               ^
           Example56.java:26: error: orphaned default
               default :e=0;
                   ^
4 errors
Compilation failed.
```

Q57.

```
import java.util.*;
class Example57 {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter integer number : ");
        int num = input.nextInt();
        for(int i = 1; i<=num;i++){
            if(num%i==0){
                System.out.print(i+" ");
            }
        }
    }
}
```

Output.



```
C:\WINDOWS\SYSTEM32\cmd X + v
Enter integer number : 56
1 2 4 7 8 14 28 56

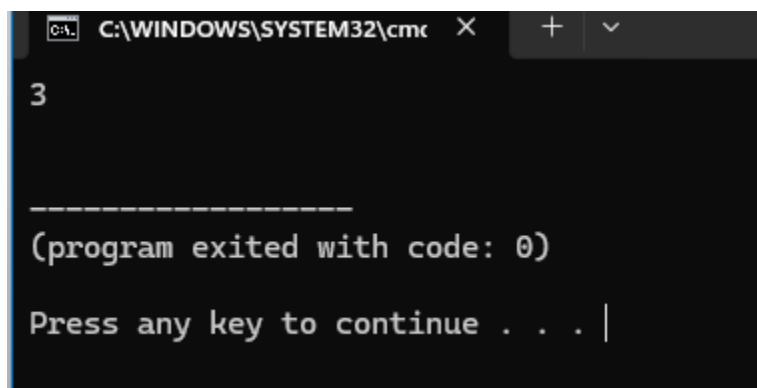
-----
(program exited with code: 0)

Press any key to continue . . .
```

Q58.

```
class Example58 {  
    public static void main(String[] args){  
        boolean b = false ;  
        int i = 1 ;  
        do{  
            i++;  
            b = !b;  
        }while(b);  
        System.out.println(i);  
    }  
}
```

Output.



A screenshot of a Windows command prompt window. The title bar shows the path 'C:\WINDOWS\SYSTEM32\cmd' and standard window controls. The command prompt displays the output of the Java program: the number '3' on the first line, followed by a horizontal line, then '(program exited with code: 0)' on the third line, and finally 'Press any key to continue . . . |' on the fourth line.

```
C:\WINDOWS\SYSTEM32\cmd X + v  
3  
  
-----  
(program exited with code: 0)  
Press any key to continue . . . |
```

Q59.

Output.

??

- A. The program will fail to compile.
- B. The program will print 3, 3 when run.
- C. The program will print 4, 3 when run if break is replaced by continue.
- D. The program will fail to compile if break is replaced by return.
- E. The program will fail to compile if break is simply removed

Answer – no one match.

Q60.

60. Which statements are true? Select the two correct answers.

- A. `{{}}` is a valid statement block.
- B. `{ continue; }` is a valid statement block.
- C. `block: { break block; }` is a valid statement block.
- D. `block: { continue block; }` is a valid statement block.
- E. The `break` statement can only be used in a loop (`while`, `do-while` or `for`) or a `switch` statement.

Answer – A and C.

Q61.

```
Example65.java  Example64.java  Example63.java  Example62.java
class Example61 {
    public static void main(String args[]){
        for(int i=0; i<10; i++){
            int j=i%3;
            switch(j){
                case 0: System.out.print("A ");
                    break;
                case 1: System.out.print("B ");
                    break;
                default: System.out.print(j+" ");
            }
        }
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd
A B 2 A B 2 A B 2 A
-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q62.

```
class Example62 {  
    public static void main (String[] args) {  
        outer:for (int i = 0; i < 3; i++) {  
            for (int j = 0; j < 2; j++) {  
                if (i == j) {  
                    continue outer;  
                }  
                System.out.println("i=" + i + ", j=" + j);  
            }  
        }  
    }  
}
```

Output.

```
i=1, j=0  
i=2, j=0  
i=2, j=1  
  
-----  
(program exited with code: 0)
```

Q63.

```
Example65.java  X  Example64.java  X  Example63.java  X  Exan
class Example63 {
    public static void main (String args[]) {
        int x=1;
        L1:if(x>0){
            System.out.print("A ");
            L2:if(x==1){break L1;}
            System.out.print("B ");
        }
    }
}
```

Output.

```
C:\WINDOWS\SYSTEM32\cmd  X  +  v
A
-----
(program exited with code: 0)
Press any key to continue . . . |
```

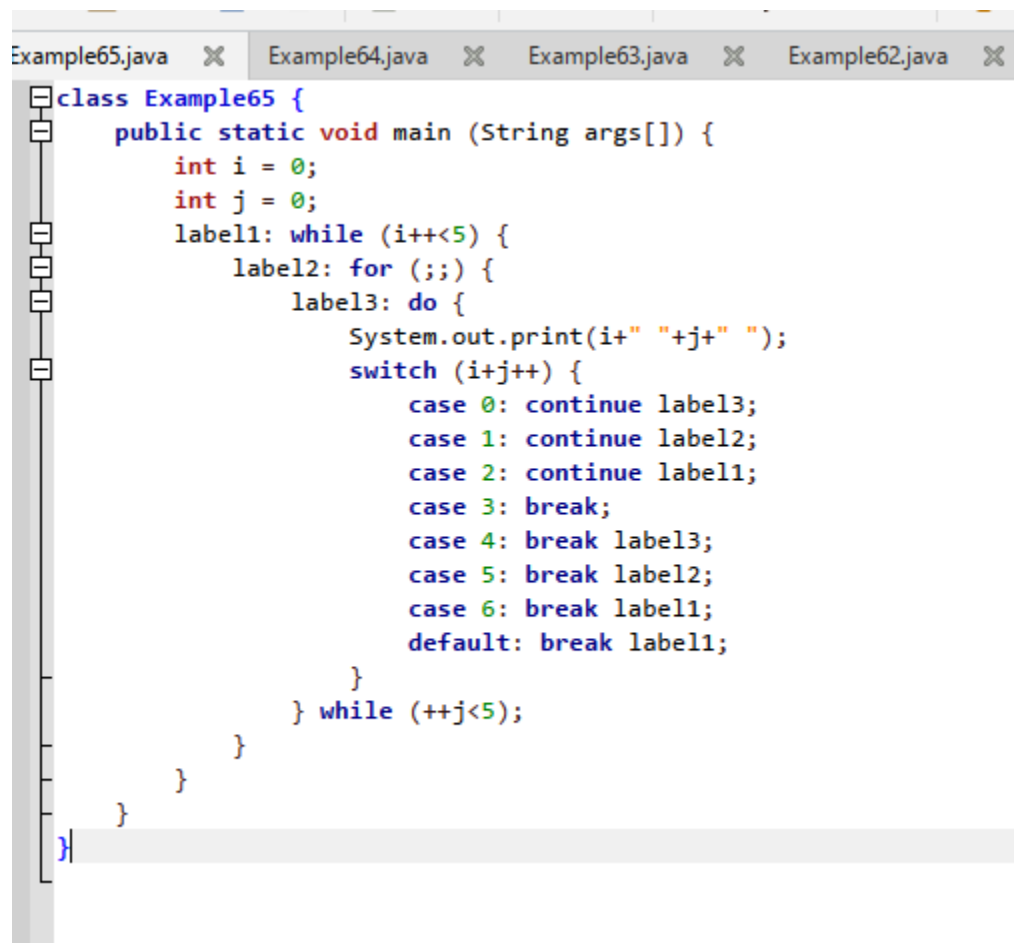
Q64.

```
Example03.java  Example04.java  Example05.java  Exa
class Example64 {
    public static void main(String args[]){
        for ( int i = 1 ; i <= 5 ; i++ ){
            if ( i < 5 ){
                continue;
            }else System.out.println(i);
        }
    }
}
```

Output.

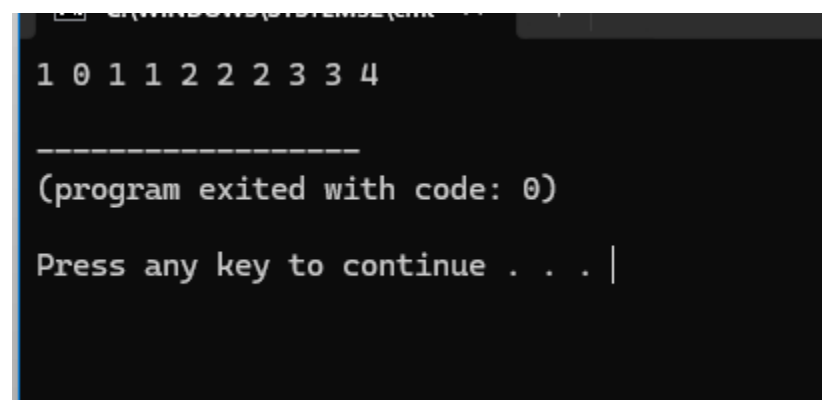
```
5
-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q65.



```
class Example65 {  
    public static void main (String args[]) {  
        int i = 0;  
        int j = 0;  
        label1: while (i++<5) {  
            label2: for (;;) {  
                label3: do {  
                    System.out.print(i+" "+j+" ");  
                    switch (i+j++) {  
                        case 0: continue label3;  
                        case 1: continue label2;  
                        case 2: continue label1;  
                        case 3: break;  
                        case 4: break label3;  
                        case 5: break label2;  
                        case 6: break label1;  
                        default: break label1;  
                    }  
                } while (++j<5);  
            }  
        }  
    }  
}
```

Output.



```
1 0 1 1 2 2 2 3 3 4  
  
-----  
(program exited with code: 0)  
  
Press any key to continue . . . |
```